Anshuman Kumar

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Skills

Programming Languages	:	C++11/14/17, Python 3.x, Rust
Scripting Languages	:	BASH, Python
Databases	:	MySQL, PostgreSQL, MongoDB, SqLite, PostGIS
GUI Frameworks	:	QT, GTK, Slint, Flutter
Build System & Tools	:	GCC, Clang suite, CMake, Meson
Infrastructure & Tools	:	Git, Docker, Sysprof, Linux, Webview
Other Libraries	:	Drogon, MapLibre GL, Matplot++, PortAudio, GTKmm
Other tools	:	Dia, Lunacy, Doxygen, Dbeaver, Postman

Work Experience

Internship at **IDR Research and Development Pvt. Ltd.** Aug 2025 - Present

1. Naviq
   - Developed a fully offline Strategic Precedent, Decision Support Engine using **Python + QT + C++**
   - Built Wargaming simulation engine using **Python** + **Ollama** + **LanceDB**
   - Architected the complete data pipeline involving **pyMuPDF, Camelot, paddleOCR**
   - Packaged and shipped the application using **Docker** for seamless cross-platform support
2. Granite
   - Built a 100% air-gapped geospatial analysis platform involving **MapLibre GL** and **PostGIS**
   - Developed **C++ based** services to power low-latency spatial indexing, event processing, offline tile rendering
   - Integrated local LLMs for **spatial clustering**, **time-series analysis** and summarization using **Python + Ollama**
3. Wi-Fi based Mesh Network
   - Built a C++-based **dynamic multiple role-based multi-node network** implemented over ESP32
   - Developed custom protocol with packet drop detection, checksum validation
   - Added **healing feature** to combat master node failures during operation

Personal Projects

1. Deck++ **Github**
   - Built a modular, DIY **Macro-Pad Ecosystem** consisting Hardware, and Customization Software
   - Designed a **4x3 matrix-based** macro-pad using an ESP32 with **ghosting prevention** and **serial communication**
   - Provided **Rust + Slint** based desktop application for controlling and customizing the macro-pad
   - Added **high-speed polling, real-time macro testing**, active-assign, while achieving ultra-low latency
2. Expenses++ **Github**
   - A minimal, high performance desktop app for financial tracking built with **Python + Slint**
   - Features dynamic income/expense logging, savings goals tracking, powered by **SQLite3**
   - Learnt **GUI Engineering** by building modular, reusable components from scratch in Slint DSL
3. SysInfo **Github**
   - A **Modern System Resource Monitor** for linux daily-users
   - Powered By **Rust**'s memory-safety guarantees and **Slint** for GUI
4. VirtAudio Cable
   - Virtual Audio router for linux written using **C++** With UI built using **Qt**
   - Enabled merging multiple audio sources into a single **virtual input** for applications with single-input limitations
   - Learnt about linux **pipes**, **streams** and **low-level audio management**

Education

B.Tech (CSE) – ITS Engineering College 2023 - Present
Intermediate – K.D.S. College 2021 - 2023
High School – DAV Public School 2021